Campus Delivery Drone Navigation System

Specification Document - Version 1.0 (Date: 2026-02-01)

1. System Overview

The Autonomous Delivery Drone performs short-range parcel deliveries across campus terrain.

2. Collision Avoidance System

Collision Avoidance System implementation is complete and integrated with forward LiDAR.

3. Routing Engine & Navigation

Dr. Vikram Rao is the owner of the Routing Engine.

The Routing Engine plans collision-free waypoints based on campus building footprints.

4. Telemetry Link Protocol

gRPC was selected as the primary telemetry protocol for low-latency ground control streaming.

5. Battery & Power Architecture

Lithium-Polymer was selected as the battery chemistry for high discharge density.

6. Optical Stabilization Notes

Note on future flight stability: we might evaluate optical flow stabilization in future revisions.

7. Payload Subsystem

Payload Gripper handles packages up to 2.5 kilograms with automatic weight sensing.

David Miller is the owner of Payload Gripper.

Status of Payload Gripper: COMPLETED.